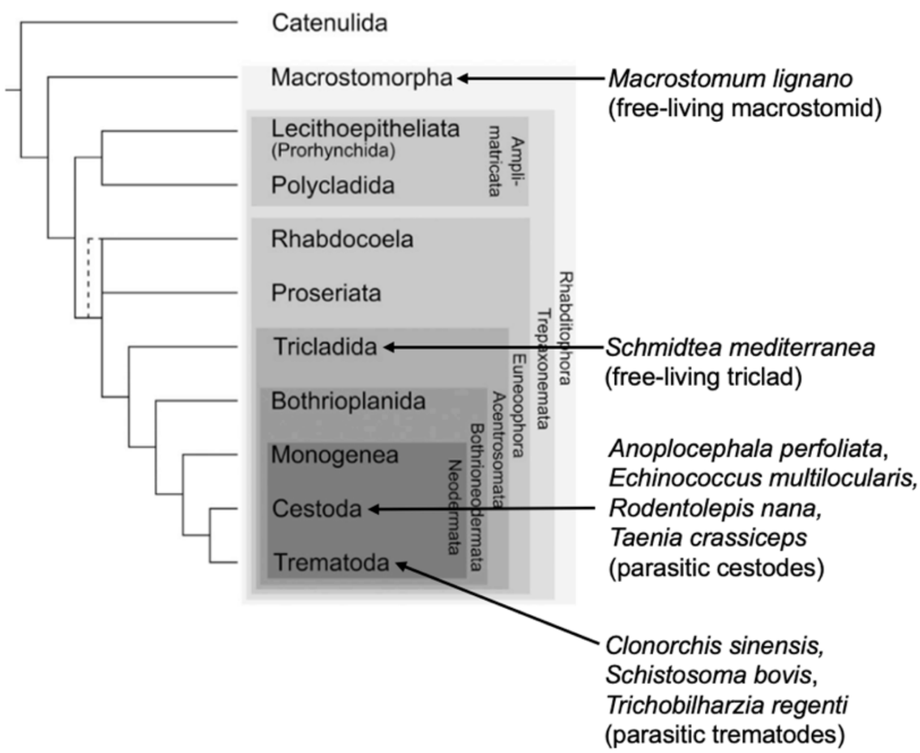
CANDIDATE CORRECTIONS TO MANUSCRIPT

- [Page 4]:
    - Figure 1: Phylogeny of metazoan species from which **HIF pathway gene** anchor sequences etc.

      Protein sequences from 25 species across 13 animal phyla were used to construct **p-HMMs for HIF-α, EGLN and VHL ortholog mining** across the proteomes of 9 platyhelminth species.

      Priapulida clades with Ecdysozoa, not Lophotrochozoa.

- [Page 5]:
    - Figure 2: **AMENDED TO**



- [Page 10]: **FINAL PARAGRAPH AMENDED TO**

  ATAC-seq consensus peakset construction
  ATAC-seq consensus peakset construction was performed using DiffBind (Stark & Brown, 2011). A consensus peakset was subsequently generated using dba.count with minOverlap = 2, retaining peaks present in at least two libraries. Peak summits were re-centred using a 250 bp summit extension.

- [Page 14]:
    - **For HIF-α ortholog mining**, alignment using MAFFT iterative local-pair refinement was selected to maximise positional homology etc.

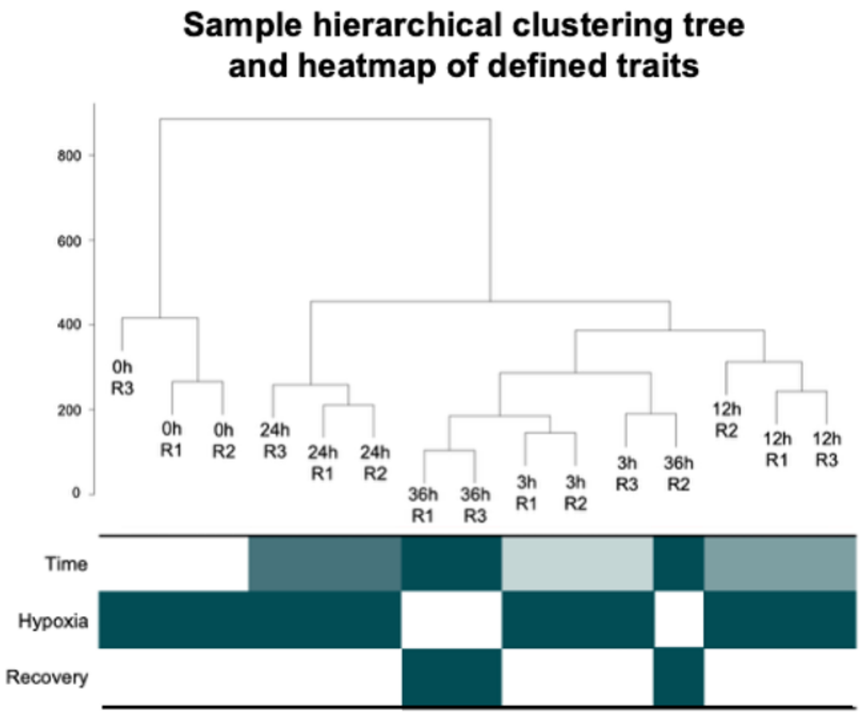
- [Page 16]:
    - A single candidate VHL ortholog in the **trematode** *Clonorchis sinensis*.

    - Subsequent maximum-likelihood phylogenetic inference placed the *Clonorchis* sequence **in a phylogenetic tree of high-confidence metazoan VHL sequences** as an isolated branch outside other metazoan VHL clades etc.

- Characteristic residue analysis showed a conserved central domain (positions 36-60) shared between the *Clonorchis* sequence and bivalve VHL-like proteins despite high divergence in flanking regions, supporting distant homology localised to a constrained functional motif **within the VHL β-domain**.

- [Page 18]:
  - Given that no orthologous sequence to the *Clonorchis* candidate VHL sequence could be identified by our HMMer search or forward BLASTp searches within the proteomes of the more basal-branching **cestodes**, *Schmidtea mediterranea* and *Macrostomum lignano*, these results suggest independent losses of VHL from **Cestoda**, Macrostomorpha and Tricladida at minimum within Platyhelminthes.

- [Page 21]:
  - Figure 11A: **AMENDED**



  - Figure 11C: Heatmaps **beside** the module assignments etc.

  - Figure 11D: Temporal trajectories of module expression showing mean FPKM **value** of genes etc.

- [Page 31]: These indirect sources of evidence are critical in a non-model metazoan with no transgenic technologies available, precluding the use of **CRE–reporter** constructs for directly testing the function and mapping the target genes of CREs.

- [Page 32]: Figure 18: Base-pair resolution view of hypoxia-specific Klf10/11 footprint in **h1SMcG0021003**.